R108/B_density_ratio | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 30 | update 43200 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0617	66.2037	11.417242	11.427978	11.371692	-0.1937	+0.3938	0.4064
TEST ALL	71.2173	66.1802	11.374505	11.385299	11.329932	-0.1938	+0.3894	0.3991

Dataset	n	NSS Top1 (%)	R108 Top1 (%)	NSS cost	R108 cost
TSP TEST	4500	45.888889	46.222222	7.970400719	7.972072475
CVRP TEST	4500	45.977778	46.622222	18.004802368	18.020025838
TSPLIB	49	20.408163	30.612245	8.217907074	8.207320926
CVRPLIB	100	70.000000	59.000000	67.101111717	67.330566254

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: half positive softplus plus half prior-weighted product-noise softplus

R108/B_density_ratio | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0617	66.2037	11.417242	11.427978	11.371692	-0.1937	+0.3938	0.4064	per-problem TRAIN SBS
TSP	46.1778	33.1111	8.026102	8.045354	7.990533	-0.2393	+0.4451	0.4334	T2T
CVRP	46.5333	18.7111	18.169123	18.269377	18.064087	-0.5488	+0.5815	0.7924	RELD_CVRP
ATSP	78.0889	47.4889	1.637702	1.676905	1.632471	-2.3378	+0.3205	0.3184	UNICO_MatPOENet
OVRP	40.9778	35.7556	7.461311	7.469579	7.415777	-0.1107	+0.6140	0.6244	MoSES_RF
VRPB	81.0667	80.8444	9.356488	9.354876	9.318750	+0.0172	+0.4050	0.4136	reld_mt
VRPL	37.3778	33.0889	12.431929	12.437959	12.383444	-0.0485	+0.3915	0.3992	MoSES_RF
VRPTW	80.0222	79.9556	18.820641	18.819431	18.766139	+0.0064	+0.2904	0.3091	reld_mt
OVRPTW	83.8667	83.7778	10.142586	10.143730	10.116859	-0.0113	+0.2543	0.2522	reld_mt
OVRPB	88.5556	88.5111	6.357085	6.357254	6.341078	-0.0027	+0.2524	0.2623	reld_mt
OVRPL	41.0667	35.9111	7.437124	7.445709	7.390416	-0.1153	+0.6320	0.6440	MoSES_RF
VRPBL	78.8222	78.5111	9.368850	9.371523	9.328366	-0.0285	+0.4340	0.4397	reld_mt
VRPBTW	78.4667	78.4222	19.947195	19.950430	19.851084	-0.0162	+0.4842	0.4650	reld_mt
VRPLTW	78.6667	78.7556	18.764975	18.765871	18.707843	-0.0048	+0.3054	0.3204	reld_mt
OVRPBL	89.3556	89.2000	6.368922	6.370093	6.352824	-0.0184	+0.2534	0.2641	reld_mt
OVRPBTW	83.6000	83.4444	10.660184	10.661098	10.623245	-0.0086	+0.3477	0.3208	reld_mt
OVRPLTW	83.5778	83.4889	10.157628	10.157423	10.129818	+0.0020	+0.2745	0.2845	reld_mt
VRPBLTW	79.6444	79.4000	19.792440	19.797044	19.706420	-0.0233	+0.4365	0.4268	reld_mt
OVRPBLTW	83.2444	83.2889	10.610077	10.609955	10.571302	+0.0011	+0.3668	0.3446	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R108/B_density_ratio | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2173	66.1802	11.374505	11.385299	11.329932	-0.1938	+0.3894	0.3991	per-problem TRAIN SBS
TSP	46.2222	34.5333	7.972072	7.989621	7.937593	-0.2196	+0.4344	0.4287	T2T
CVRP	46.6222	17.9333	18.020026	18.124983	17.918509	-0.5791	+0.5665	0.7571	RELD_CVRP
ATSP	77.3556	47.2444	1.634957	1.674207	1.629441	-2.3444	+0.3385	0.3372	UNICO_MatPOENet
OVRP	41.0667	33.6444	7.445454	7.455750	7.400487	-0.1381	+0.6076	0.6145	MoSES_RF
VRPB	80.5778	80.4000	9.352085	9.351810	9.315298	+0.0029	+0.3949	0.3947	reld_mt
VRPL	37.4444	32.2667	12.350320	12.356138	12.300958	-0.0471	+0.4013	0.4099	MoSES_RF
VRPTW	81.1778	80.8444	18.688804	18.690545	18.637408	-0.0093	+0.2758	0.2872	reld_mt
OVRPTW	83.8667	83.9333	10.093105	10.093063	10.068135	+0.0004	+0.2480	0.2387	reld_mt
OVRPB	88.0889	88.1333	6.370968	6.370826	6.352225	+0.0022	+0.2951	0.3034	reld_mt
OVRPL	41.4222	34.6000	7.463111	7.472542	7.418033	-0.1262	+0.6077	0.6185	MoSES_RF
VRPBL	79.4222	79.4222	9.395983	9.396379	9.355967	-0.0042	+0.4277	0.4280	reld_mt
VRPBTW	78.8222	78.7556	19.664894	19.665725	19.577666	-0.0042	+0.4455	0.4393	reld_mt
VRPLTW	79.2889	79.2444	18.801244	18.803136	18.745143	-0.0101	+0.2993	0.3181	reld_mt
OVRPBL	89.2222	89.2889	6.376008	6.375741	6.359310	+0.0042	+0.2626	0.2742	reld_mt
OVRPBTW	84.8889	84.8222	10.660489	10.659591	10.624748	+0.0084	+0.3364	0.3089	reld_mt
OVRPLTW	82.9778	82.8667	10.085297	10.086112	10.059001	-0.0081	+0.2614	0.2611	reld_mt
VRPBLTW	79.8000	79.7333	19.704470	19.707068	19.613284	-0.0132	+0.4649	0.4480	reld_mt
OVRPBLTW	83.6444	83.5778	10.661797	10.662149	10.625574	-0.0033	+0.3409	0.3172	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R108/B_density_ratio | LIB and locked provenance

Current R108 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	30.6122	10.2041	8.207321	8.240435	8.142167	-0.4019	+0.8002	0.8270
CVRPLIB	59.0000	45.0000	67.330566	67.245364	66.945093	+0.1267	+0.5758	0.7022

SHA256: fb5ee298c95594197eb68f5b1030b020fb200fb451d6f1ea9db6df4629746874

Unique selection: strict minimum whole18 VAL macro_vs_sbs_pct; first exact tie; common full-TRAIN SBS

Inference: fixed TRAIN conditional log prior plus learned residual utility; original differences/maximin argmax; no quota or posthoc tuning

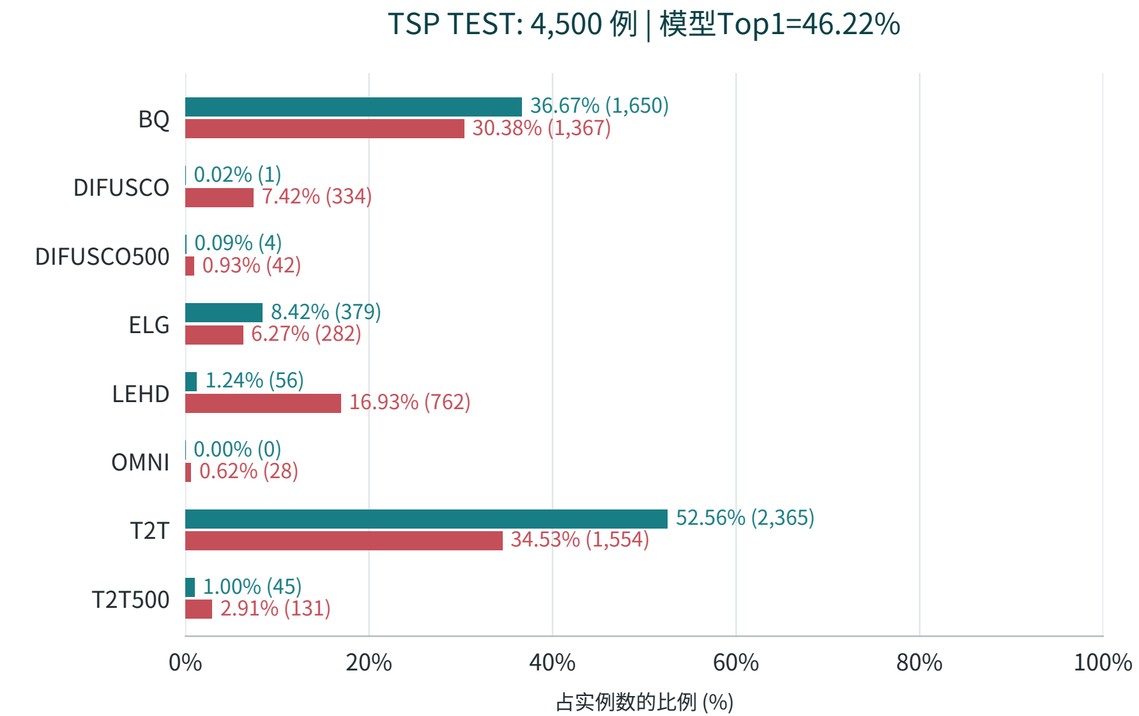
Normalization: new81000 TRAIN only; no TEST/LIB refit.

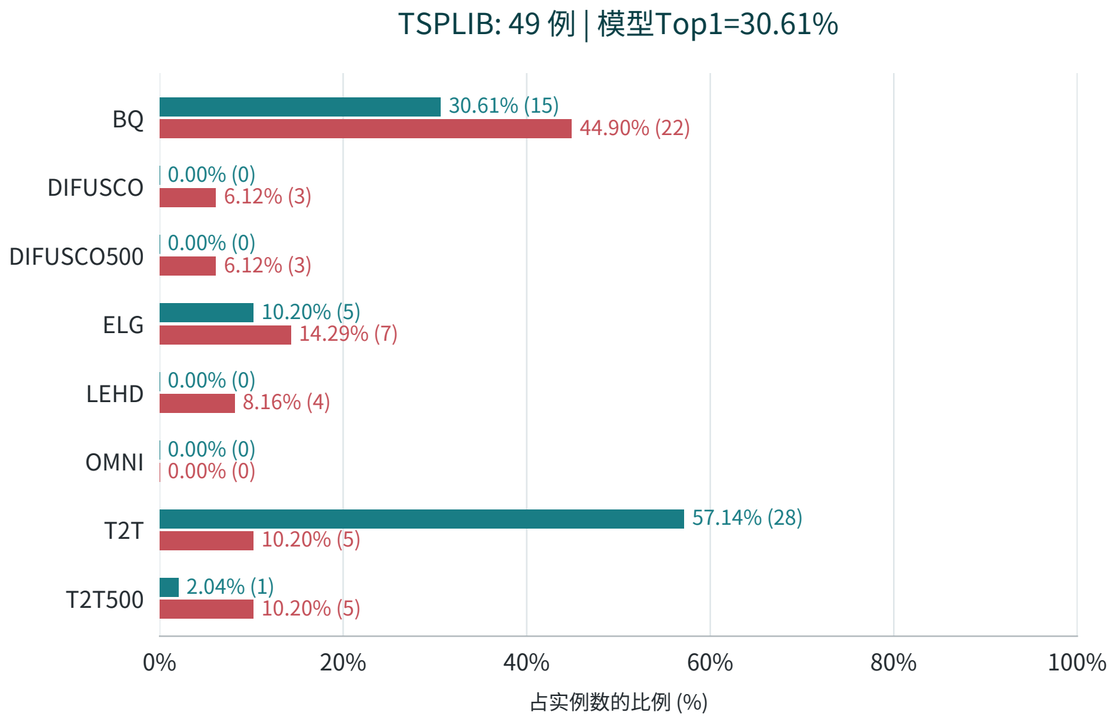
R108/B_density_ratio | TSP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

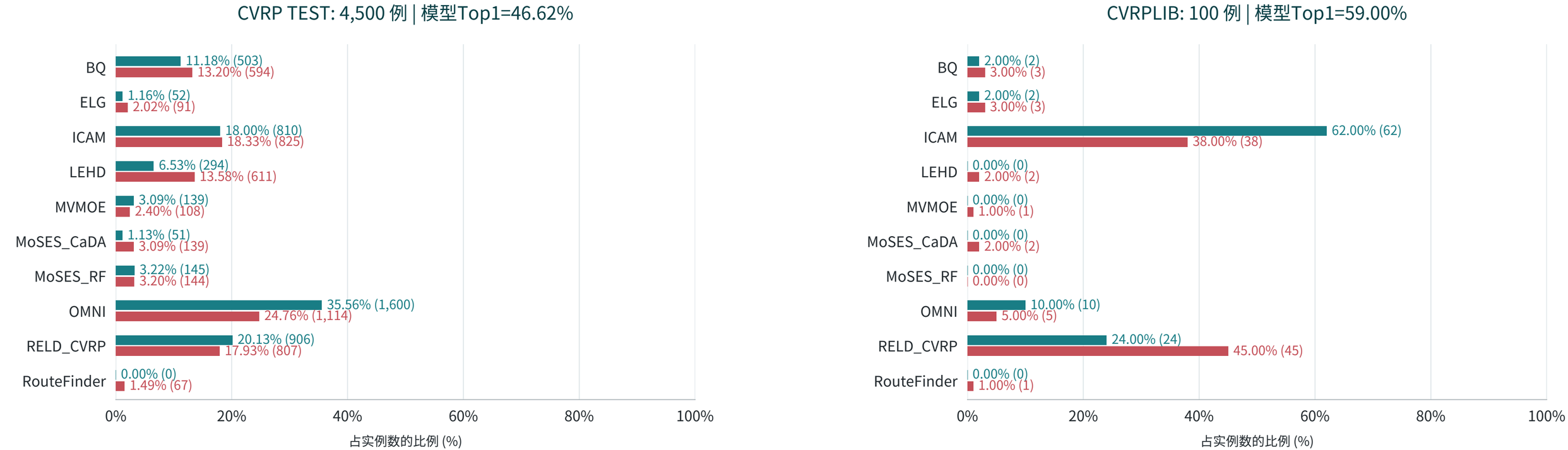
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | CVRP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

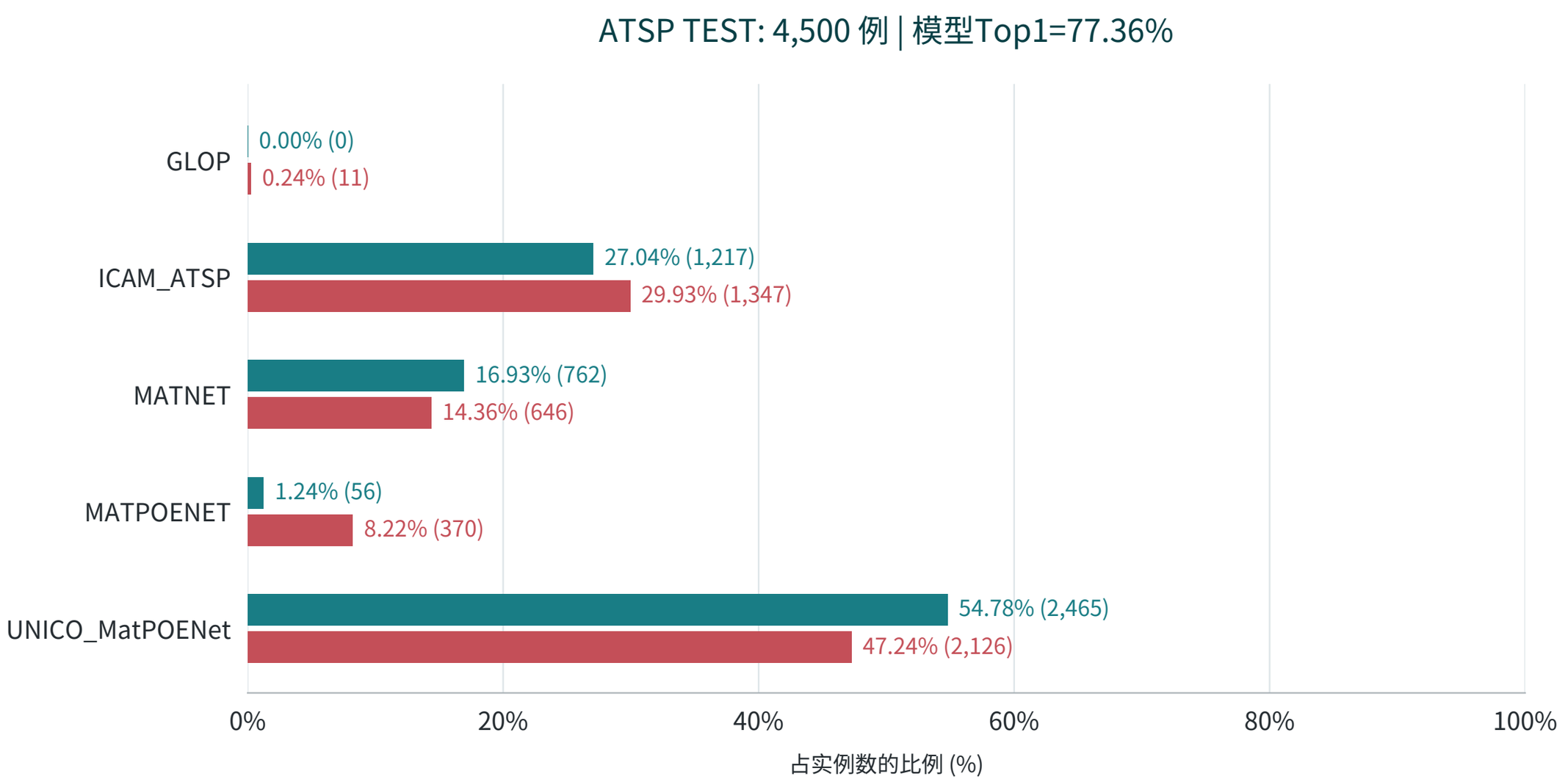
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | ATSP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

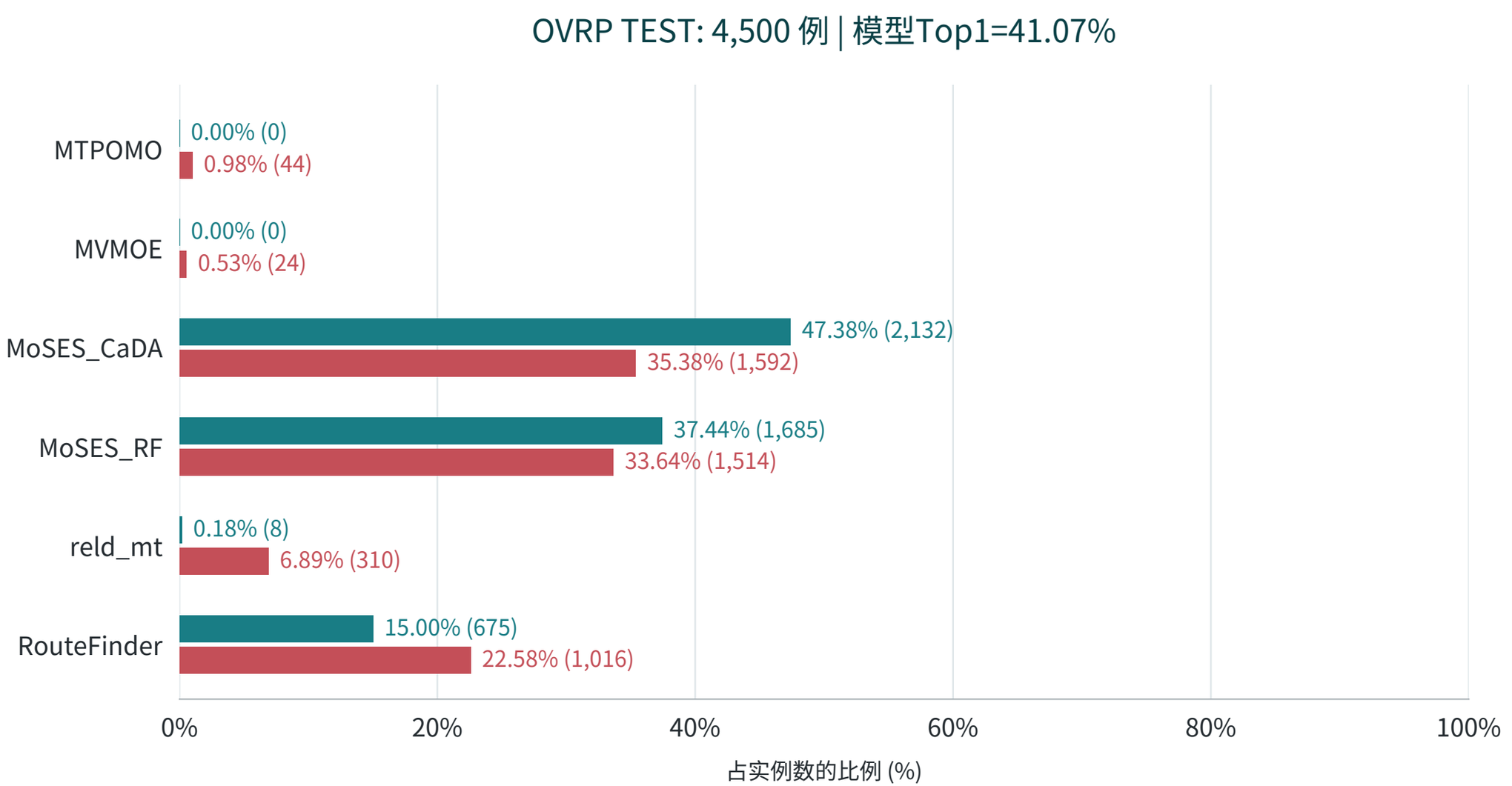
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

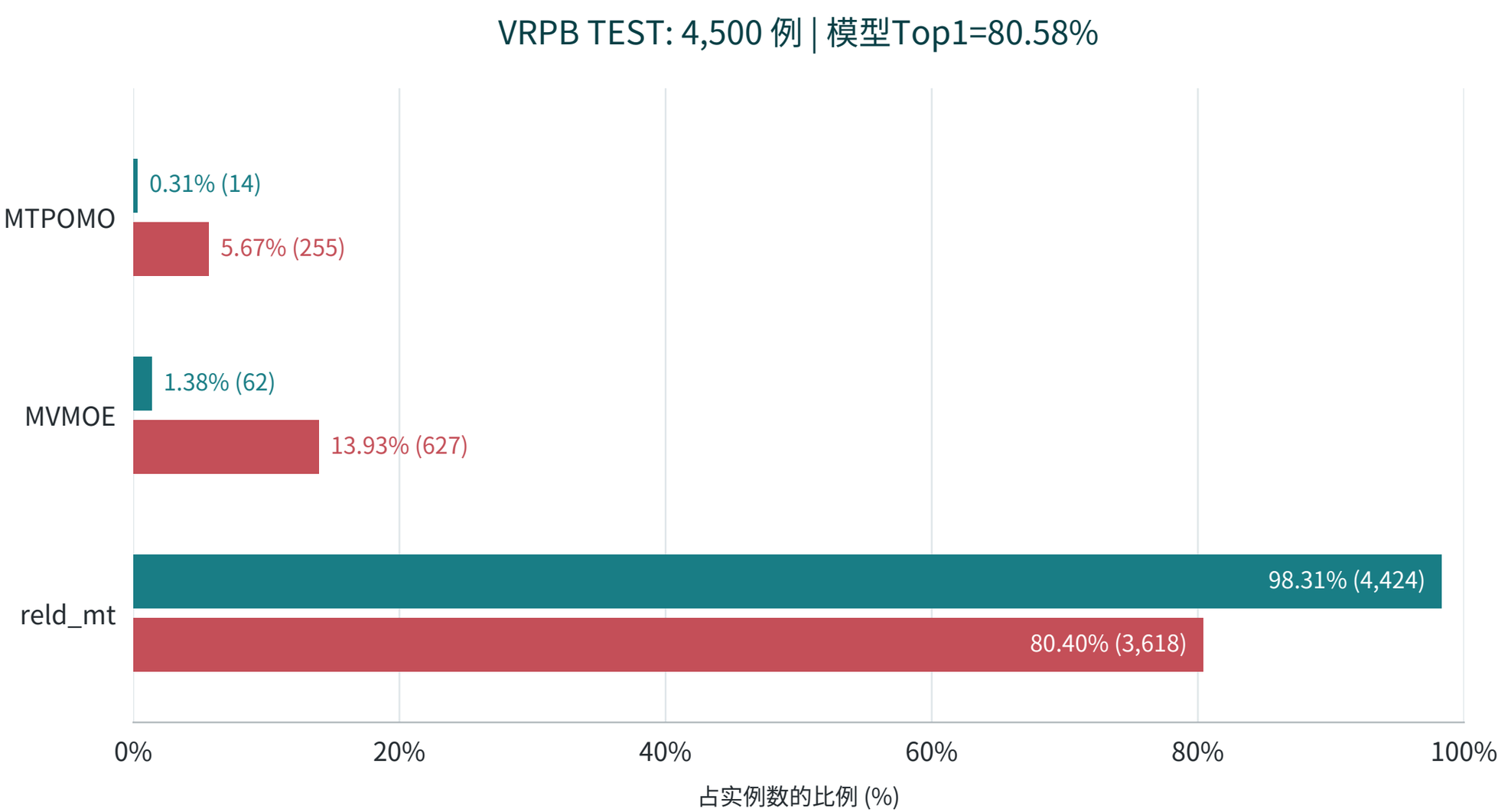
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPB | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

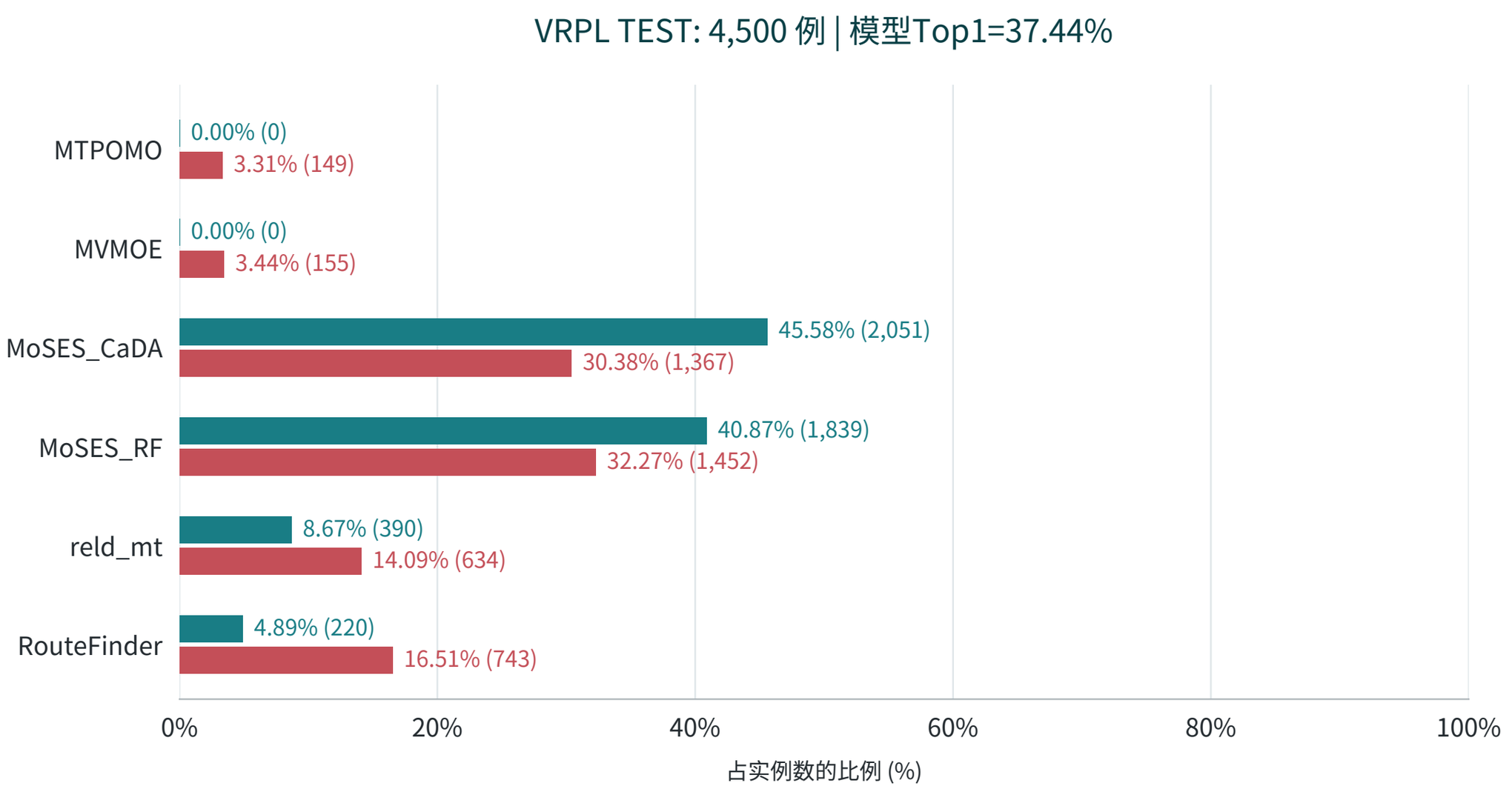
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

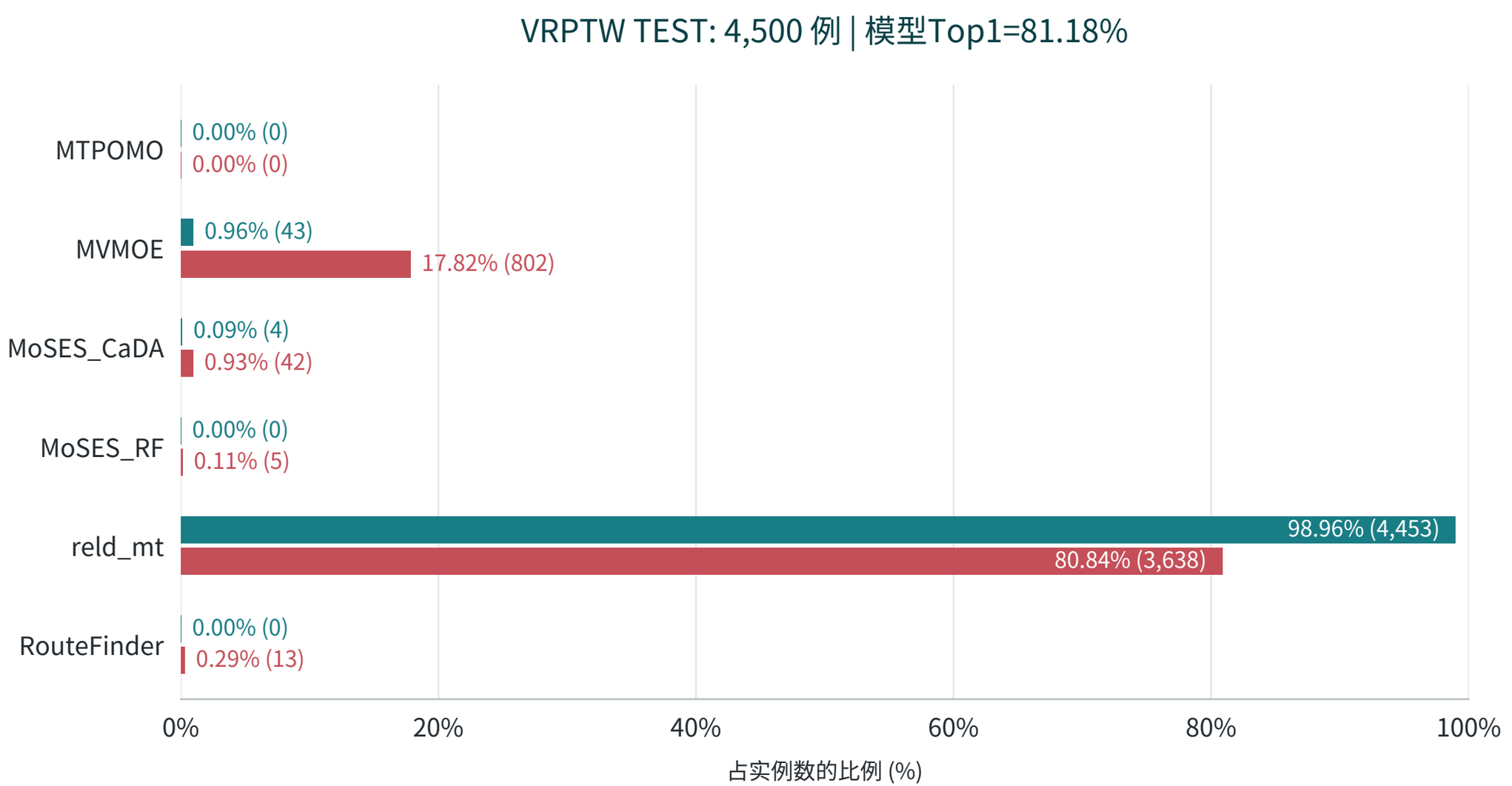
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

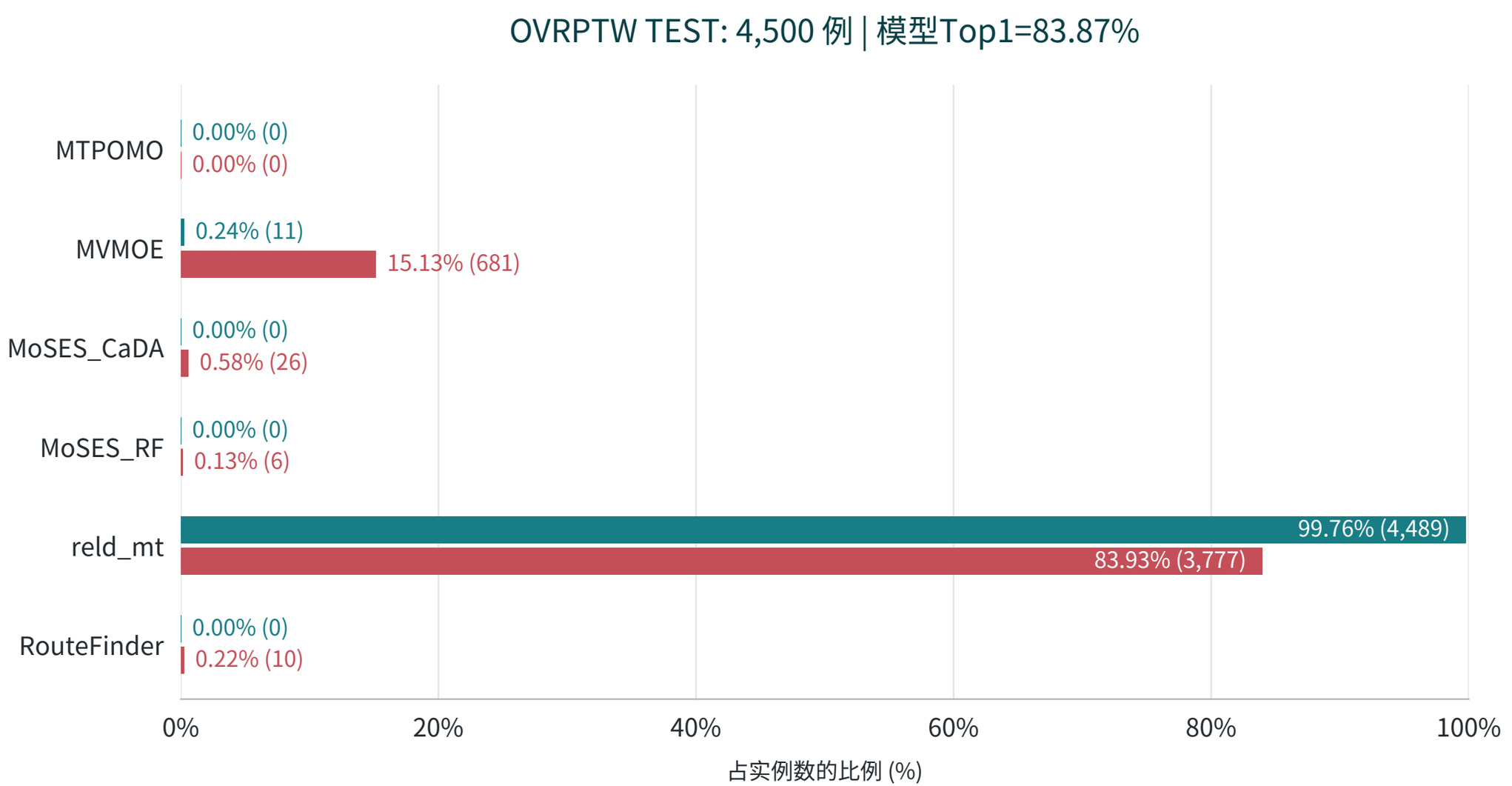
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

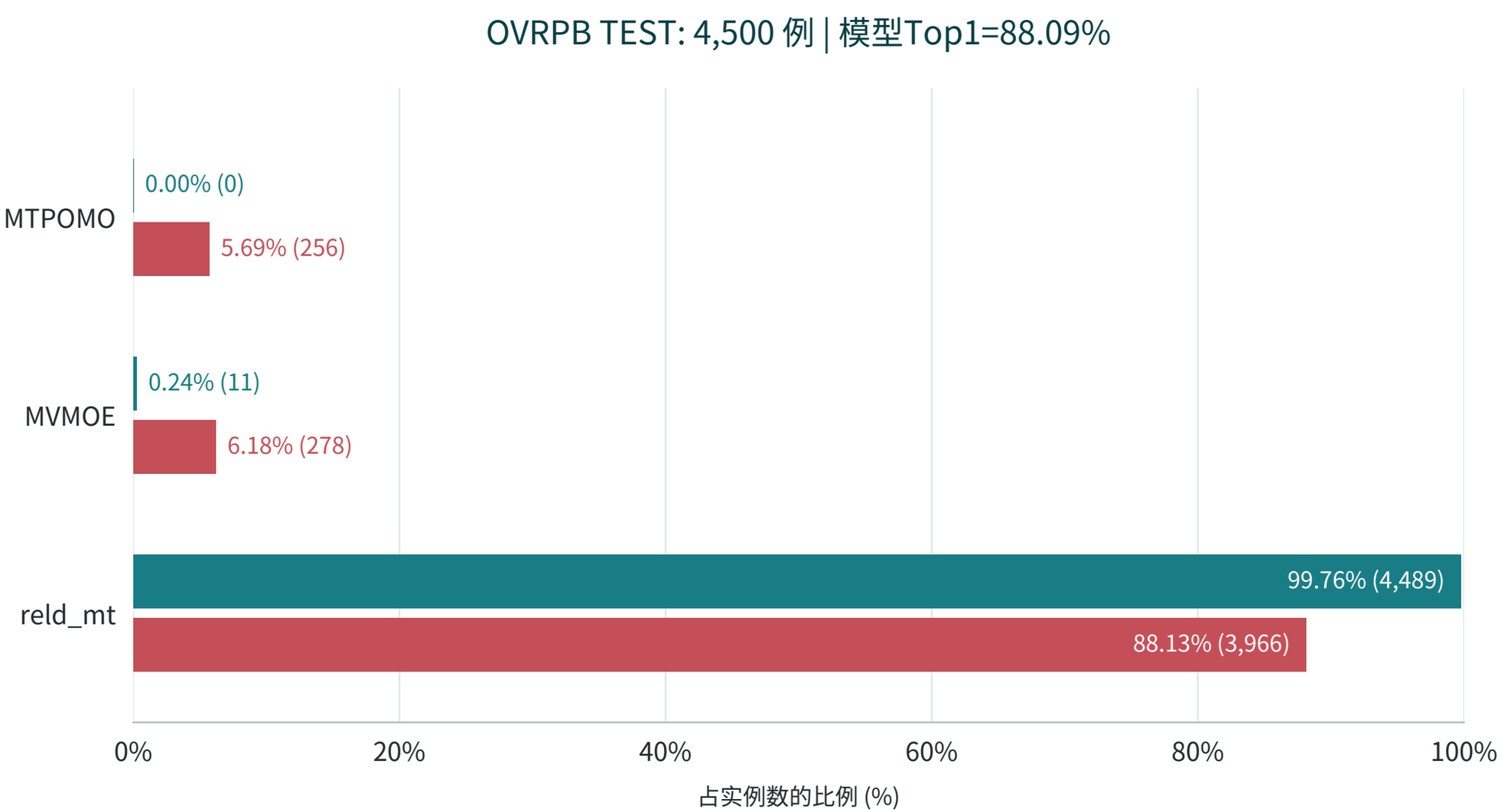
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPB | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

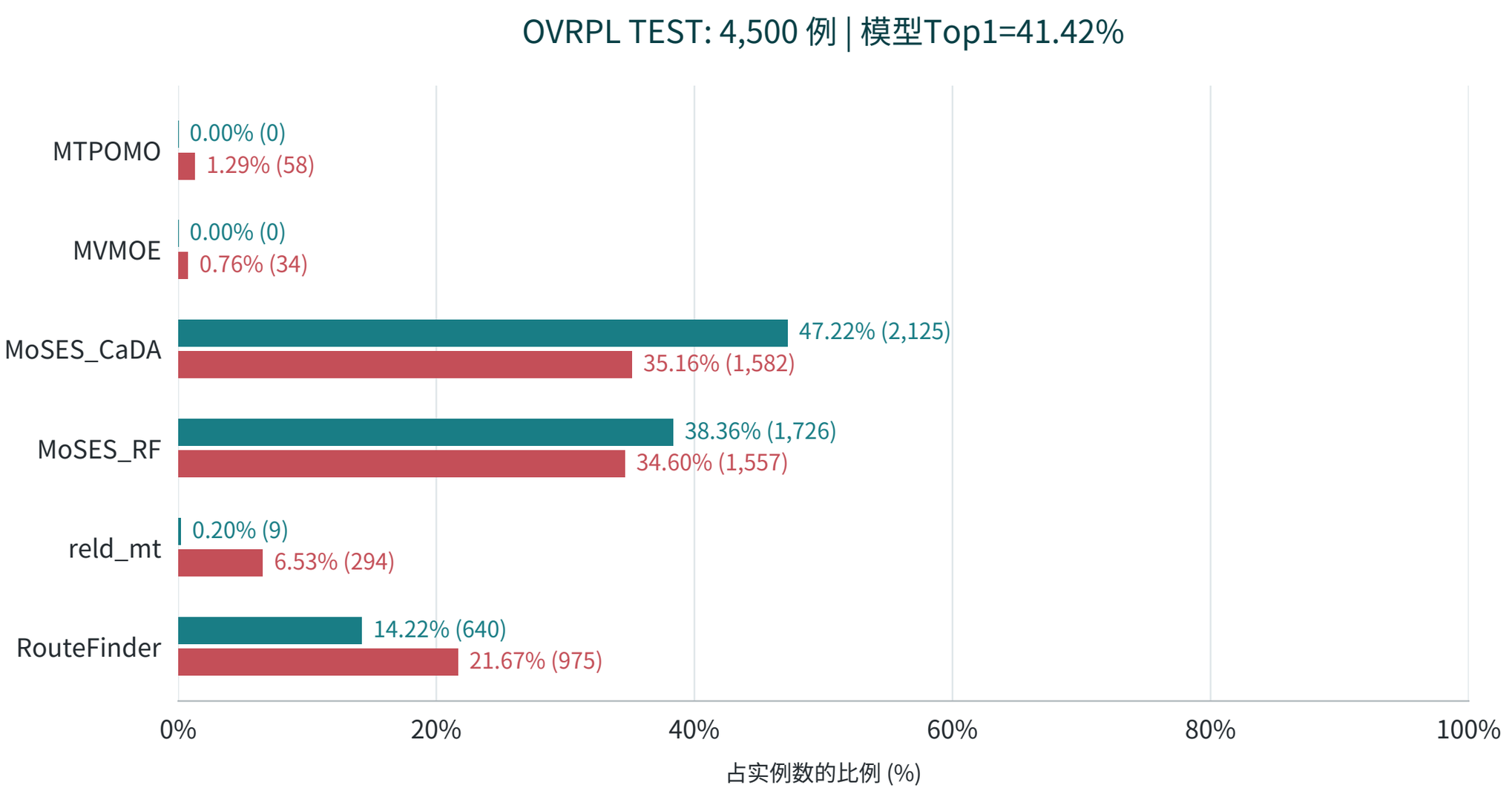
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

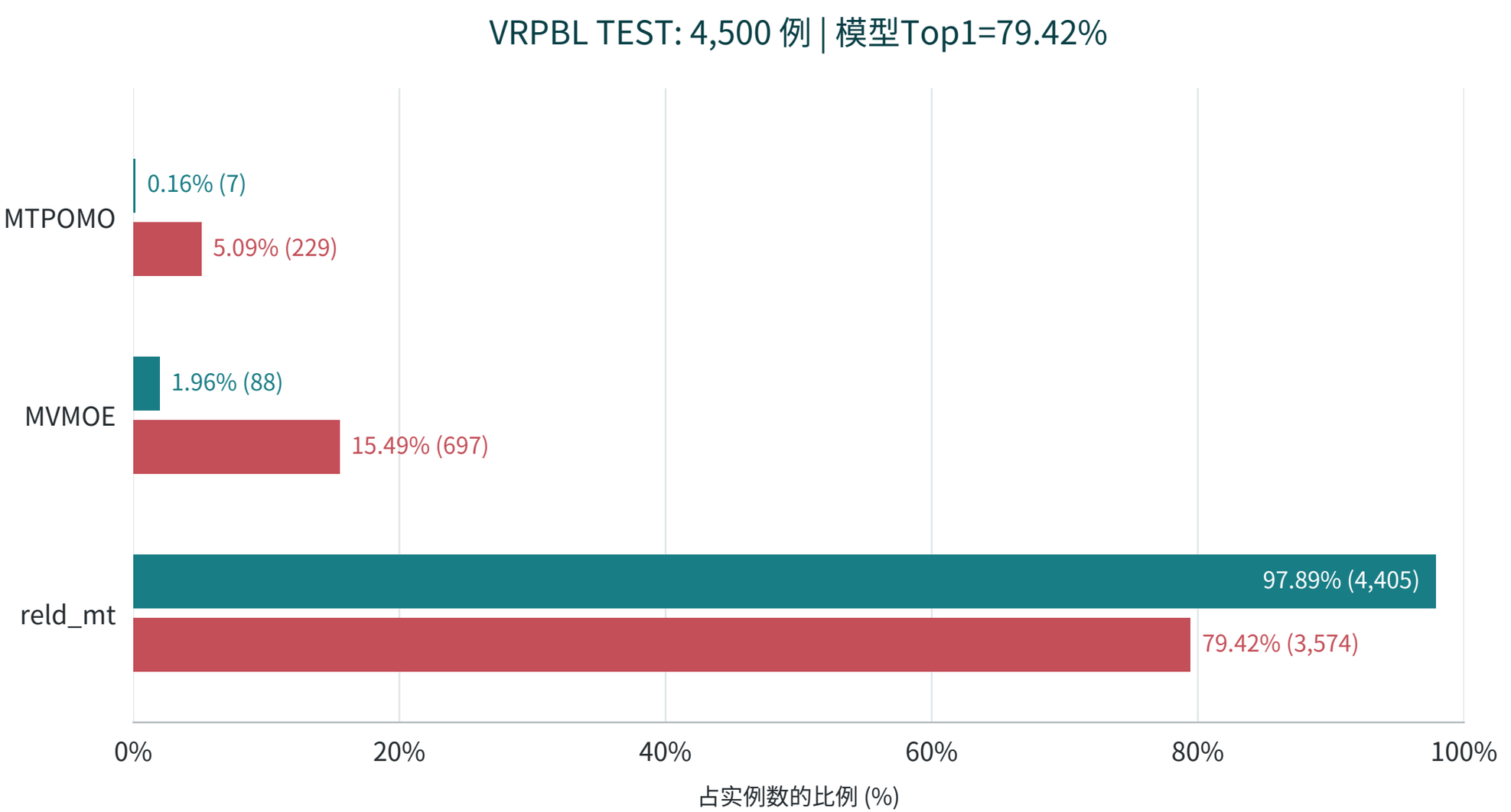
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPBL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

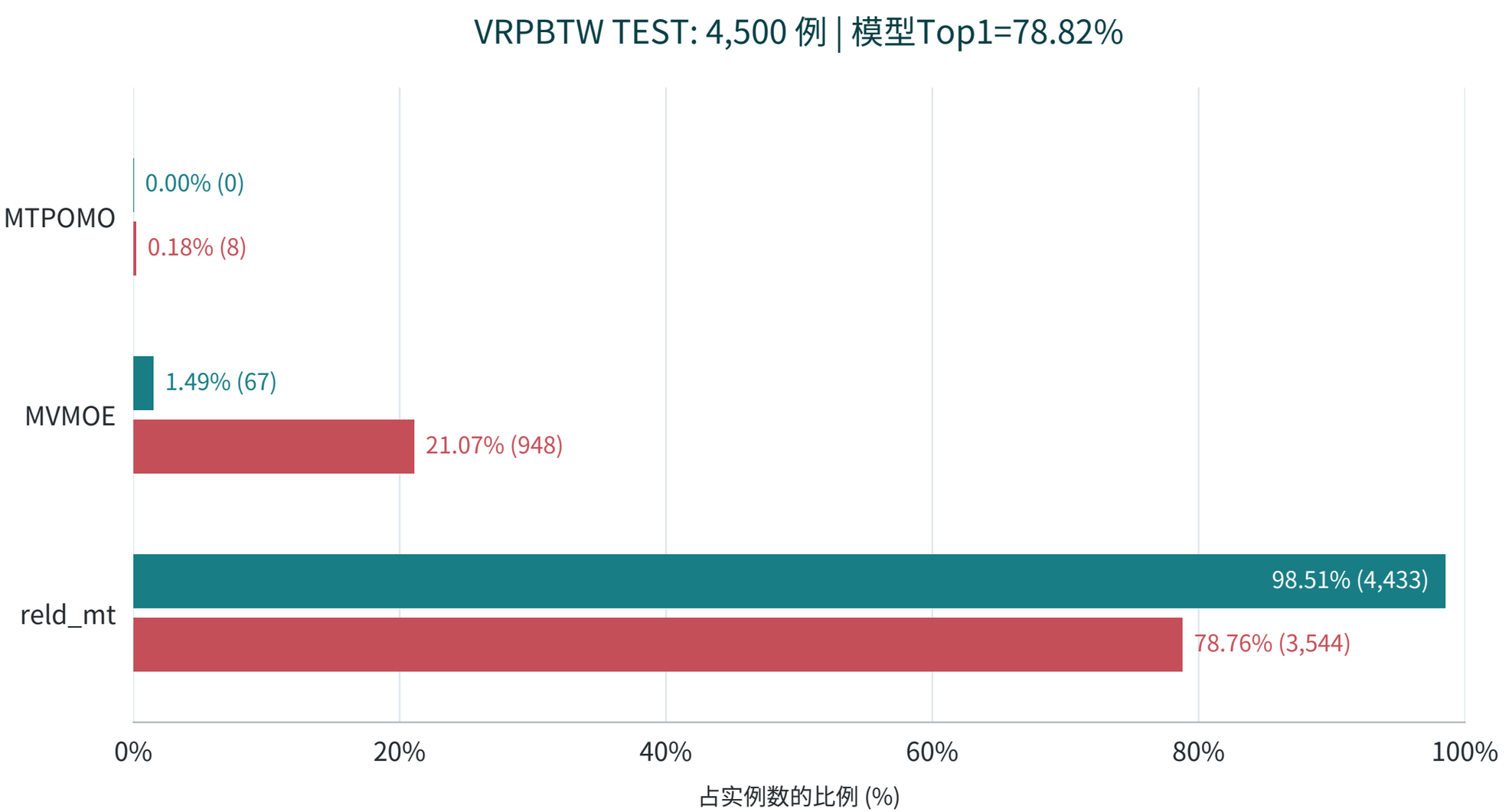
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPBTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

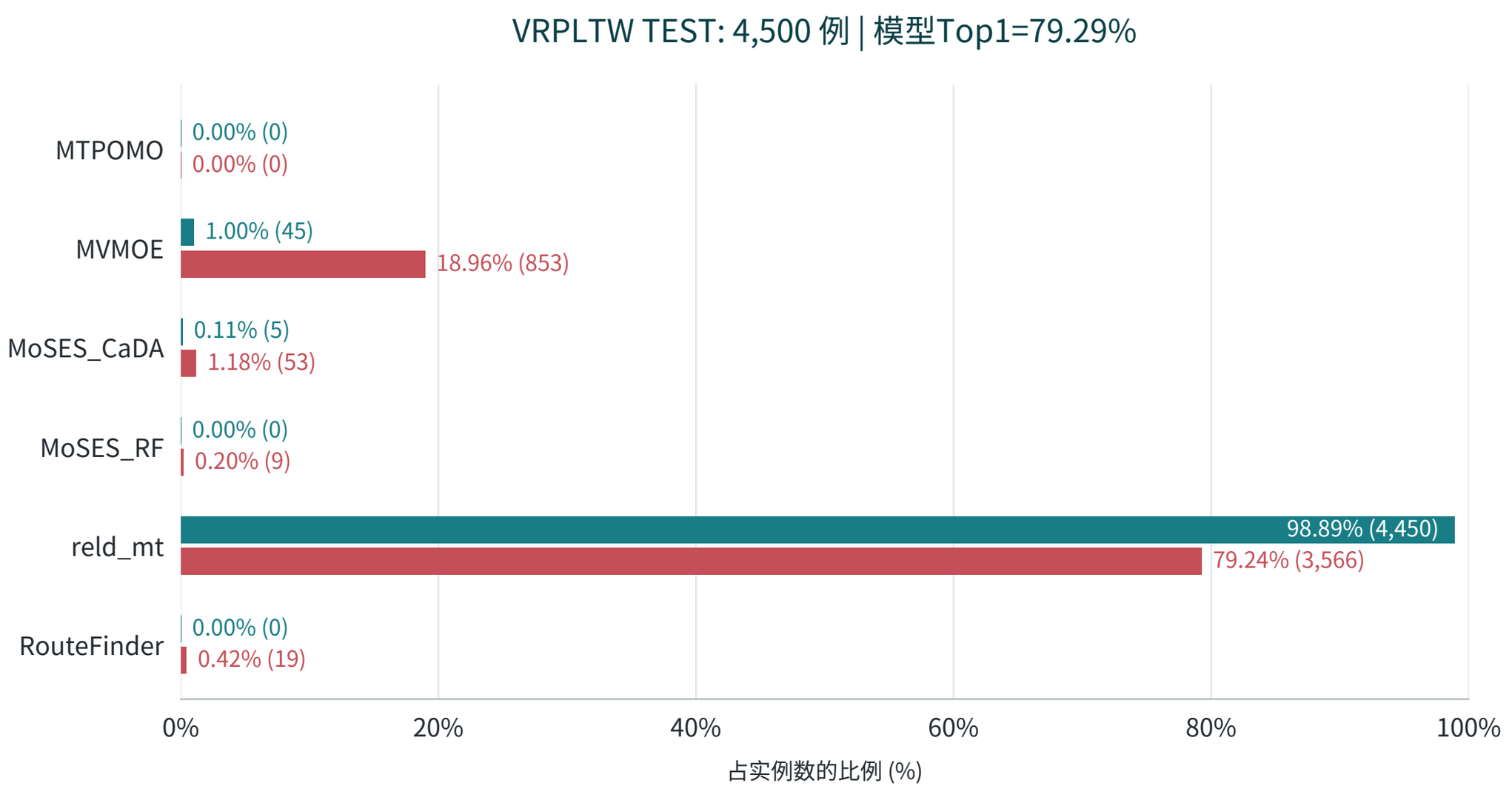
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPLTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

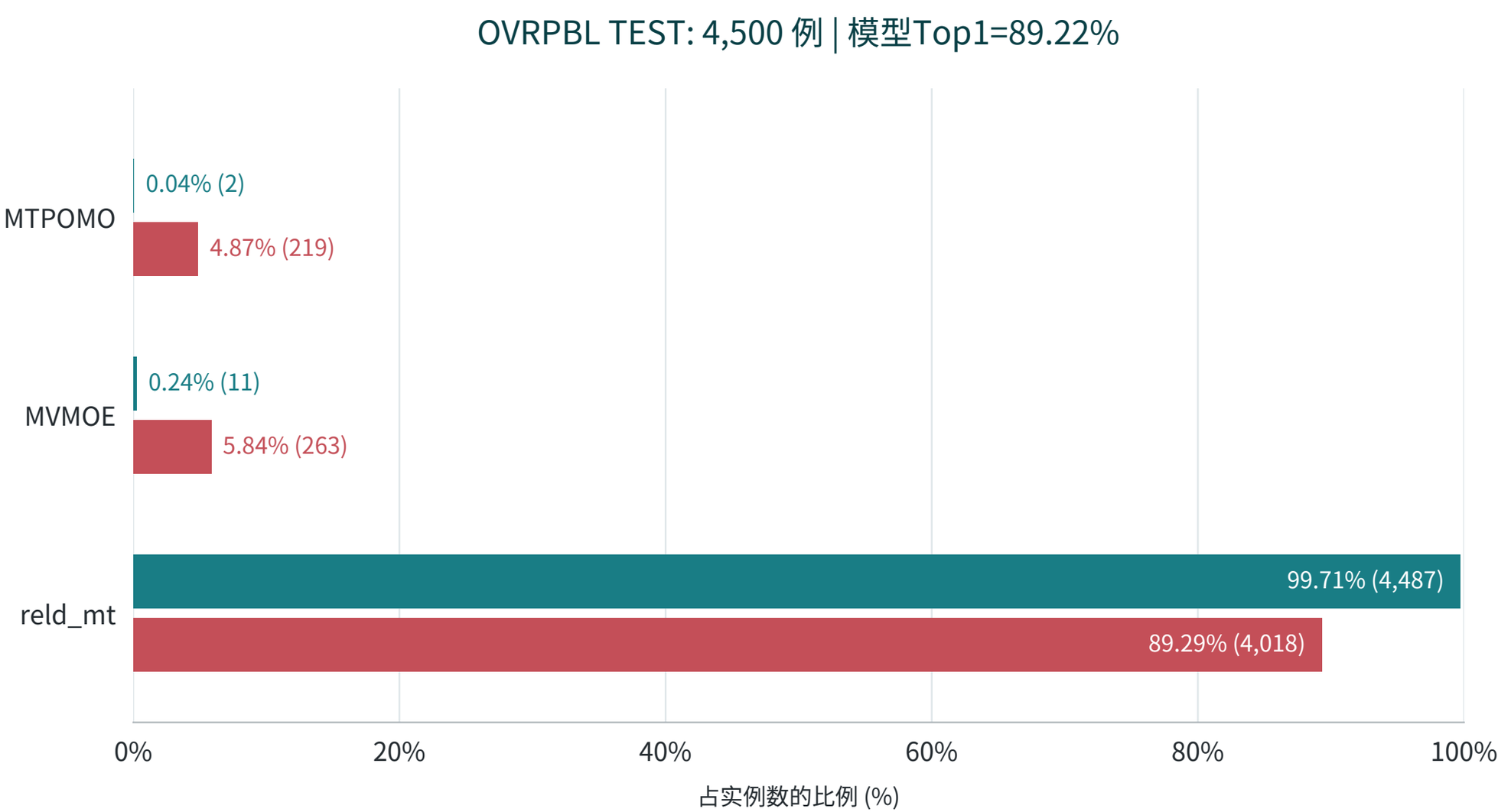
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPBL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

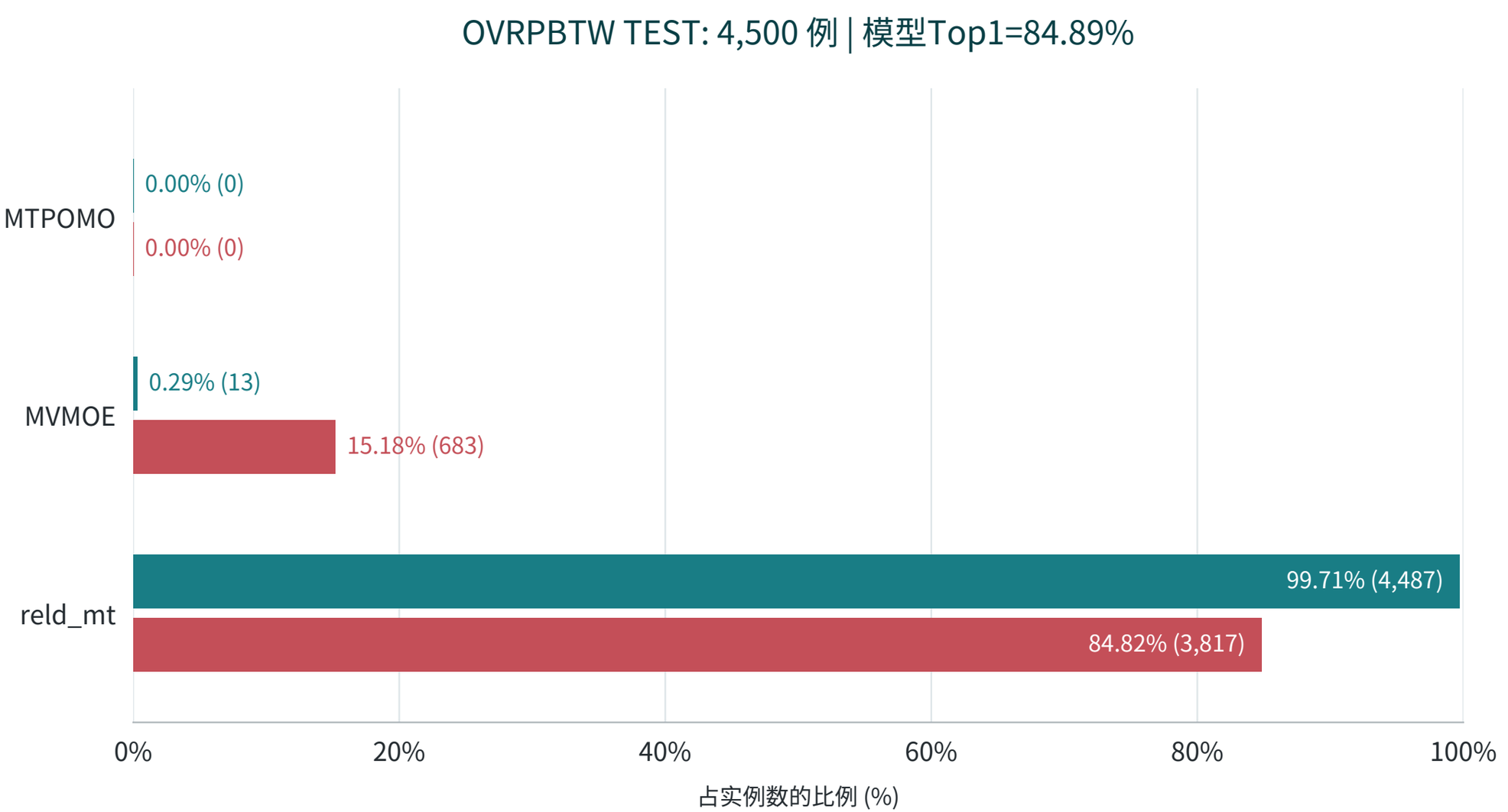
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPBTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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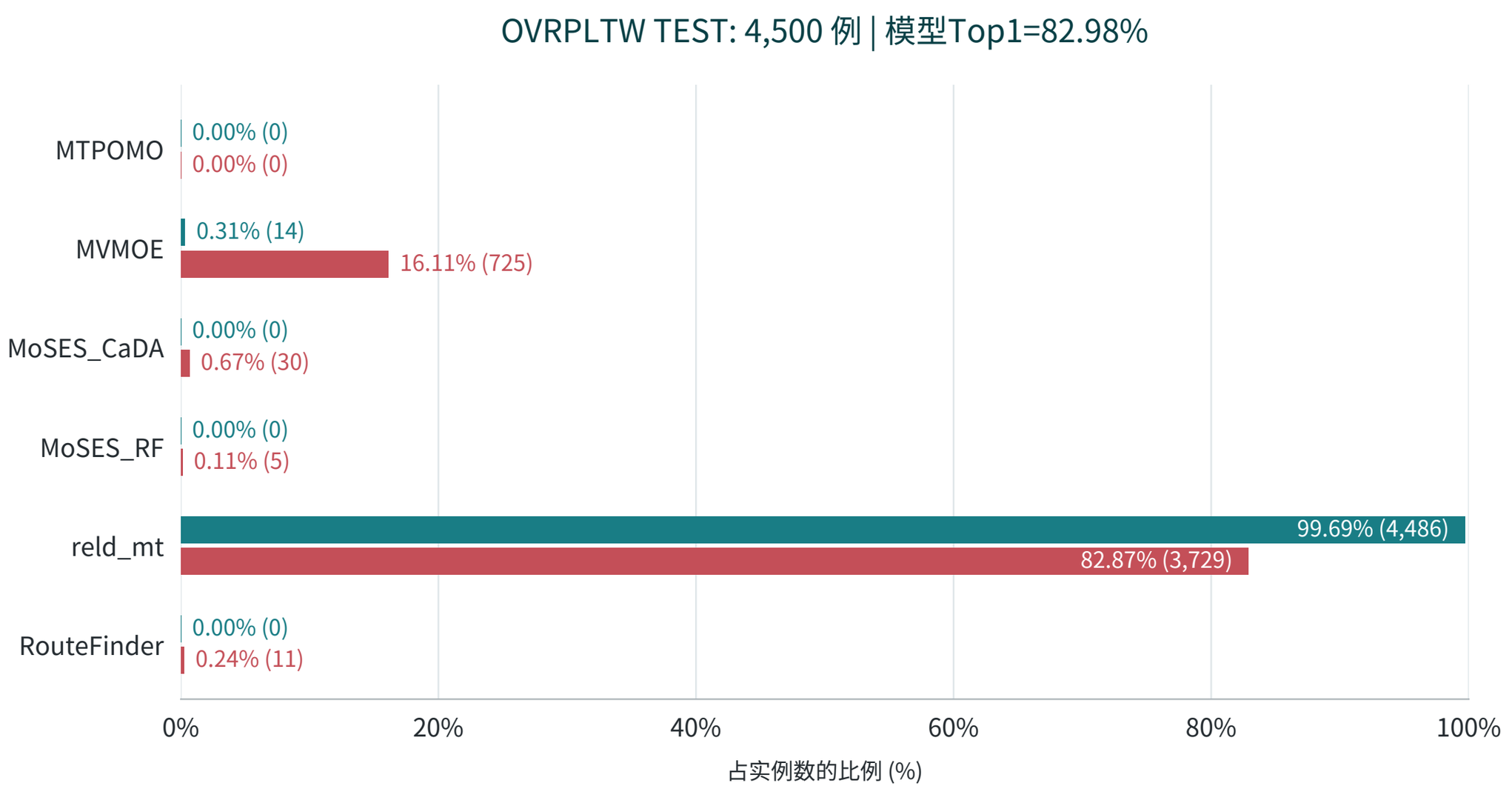
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | OVRPLTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 fb5ee298c9559419

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

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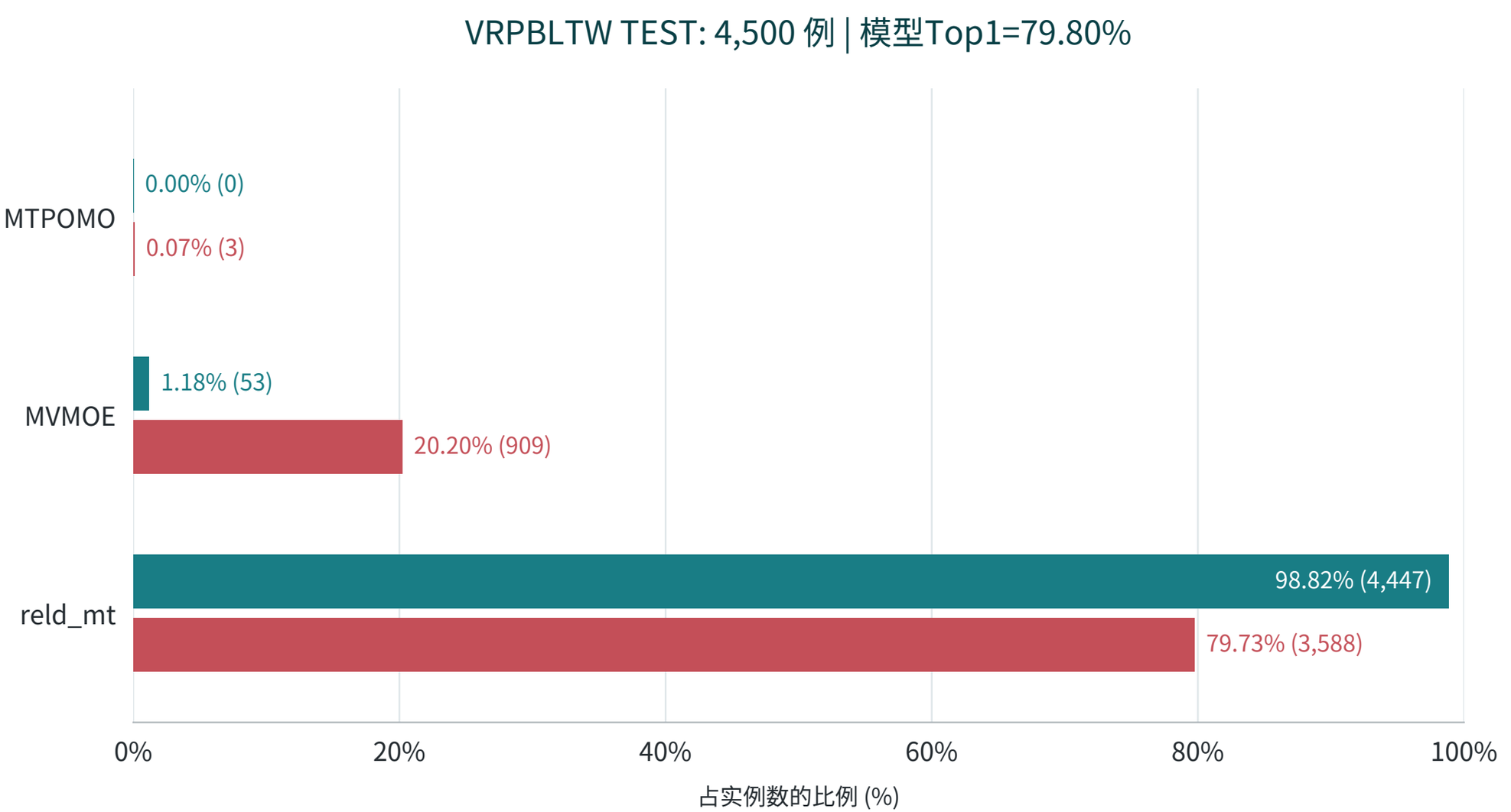
R108/B_density_ratio: stable global solver-ID score ties. Same R108 lock; LIB inference only; no training/reselection/Rxx mixing.

R108/B_density_ratio | VRPBLTW | 方法分布

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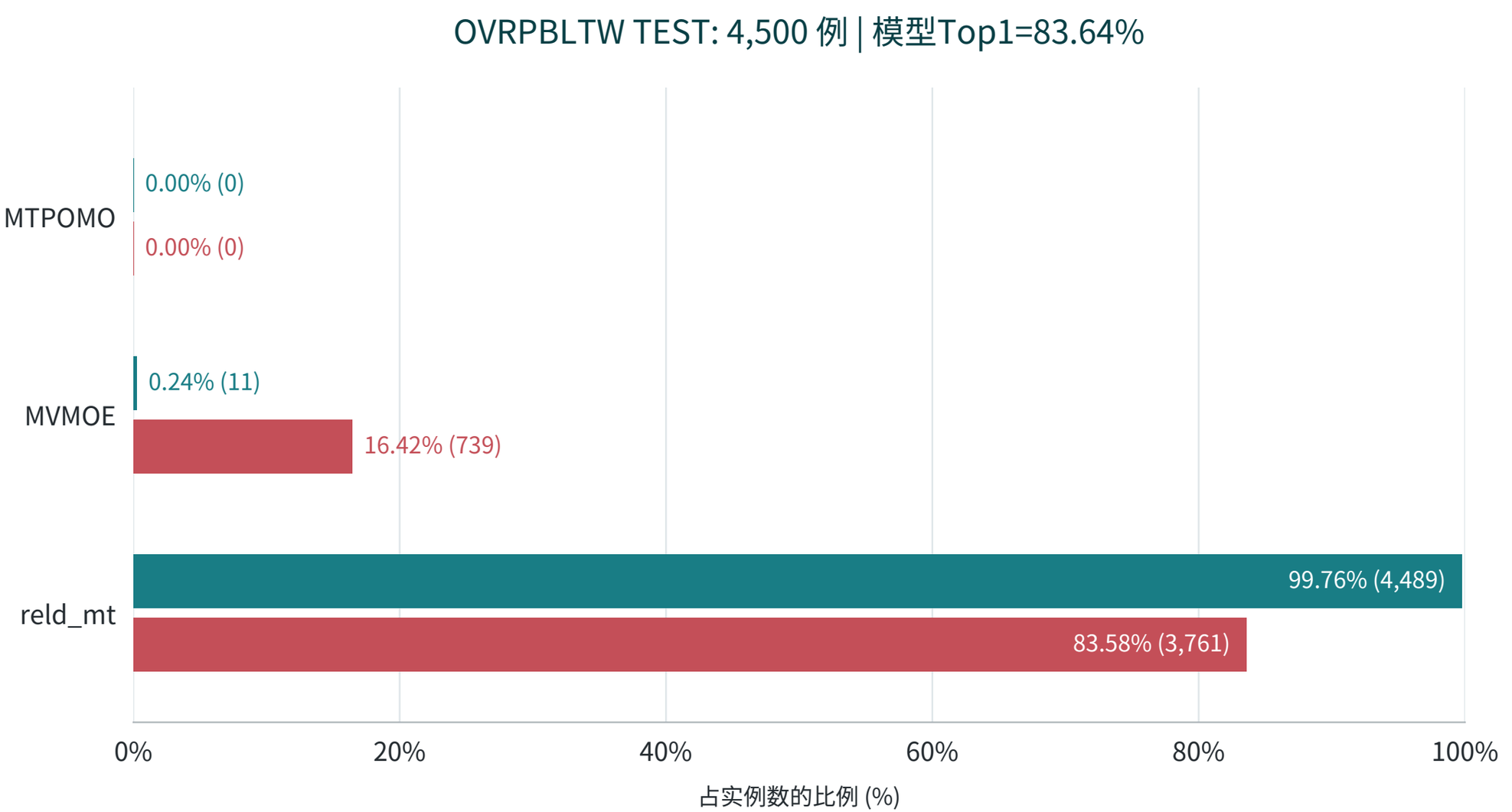
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R108/B_density_ratio | OVRPBLTW | 方法分布

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